Teaching order for

IGCSE Additional Mathematics 0606

Teachers: Mr. Leo and Mr. T

Plan

Term 1	Term 2
70% of the syllabus	30% of the syllabus

(paragraphs are according to the coursebook – New Additional Mathematics, published by Marshall Cavendish Education, ISBN 978-981-4210-18-8)

	TERM 1	
2.1—2.2	Simultaneous equations	
	Simultaneous linear equations in two unknowns, simultaneous linear and non-linear equations in two unknowns.	
3.1—3.6	Indices, surds and logarithms	
	Indices (exponents) and surds, exponential functions, logarithms, common and natural logarithms, laws of logarithms, logarithmic equations.	
4.1—4.3	Quadratic expressions and equations	
	Maximum/minimum value of a quadratic expression, roots of a quadratic equation, solving quadratic inequalities.	
5.1—5.4	Remainder and factor theorems	
	Polynomial identities, remainder theorem, factor theorem, solving cubic equations.	
7.1—7.8	Coordinate geometry	
	Distance between two points, midpoint of the line joining two points, equations of straight lines, equations of parallel and non-parallel lines, equations of perpendicular lines, perpendicular bisector, intersection of a straight line and a curve.	
8.1	Linear law	
	Linear law.	
9.1—9.4	Functions	
	Introduction to functions, composite functions, inverse functions, absolute valued functions.	
	Revision 1	
	Midterm (October)	
10.1—10.4	Trigonometric functions	2 Weeks
	Trigonometric ratios and general angles, trigonometric ratios of	

	any angle, graphs of the sine, cosine, and tangent functions, three more trigonometric functions.	
11.1—11.2	Simple trigonometric identities and equations	1 Week
	Simple identities, trigonometric equations and more graphs.	
12.1—12.2	Circular measure	1 Week
	Radian measure, arc length and area of a sector.	
13.1—13.3	Permutations and combinations	1 Week
	The basic counting principle, permutations, combinations.	
14.1—14.2	Binomial theorem	1 Week
	The binomial expansion of $(1+b)^n$, the binomial expansion of $(a+b)^n$.	
	Revision 2	1 Week
	End of term (December)	

	TERM 2	
15.1—15.5	Differentiation and its techniques	1 Week
	The gradient function, function of a function (composite function), product of two functions, quotient of two functions, equations of tangent and normal.	
16.1—16.3	Rates of change	2 Weeks
	Constant rate and variable rate of change, related rates of change, small changes.	
17.1—17.3	Higher derivatives and applications	1 Week
	Determination of maximum and minimum points, maximum and minimum points, maximum and minimum values.	
18.1	Derivatives of trigonometric functions	1 Week
	Differentiation of trigonometric functions.	
19.1—19.2	Exponential and logarithmic functions	1 Week
	Exponential functions, logarithmic functions.	
7.bis	Coordinate geometry of the circle	1 Week
	Equation of a circle, intersection of a circle and a straight line, tangents to a circle, intersections of two circles.	
20.1—20.4	Integration	1 Week
	Integration as the reverse process of differentiation and indefinite integrals, definite integrals, integration of trigonometric functions, integration of exponential functions.	
	Revision 3	1 Week
	Mock exam (March)	
21.1—21.2	Applications of integration	1 Week
	Area between a curve and an axis, area bounded by two curves.	
22.1—22.3	Kinematics	1 Week

	Displacement, velocity and acceleration, displacement-time and velocity-time graphs.	
23.1—23.4	Vectors	1 Week
	Basic concepts, vectors expressed in terms of two non-parallel vectors, position vectors, vectors in the cartesian plane.	
	Revision 4	